Abliterated Models in Appsec Pentesting

Abliterated Models in Appsec Pentesting - Laurence Tennant, Include Security.

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Content

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Intro

At Include Security we are constantly evaluating the best models to support us in webapp pentesting. Z.ai's GLM-5.3 was released in August 2026, an open weight LLM which NIST estimated was four months behind the US frontier models in cyber capabilities. Anthropic later published an article raising concerns about GLM-5.3's safeguards, particularly in abliterated versions of the model.

To recap: abliteration (a portmanteau of "ablate" and "obliteration") is a technique used to remove safety refusals from open-weight models. At a high level, abliteration edits a model's internal representations of harmful inputs to look like harmless ones. This can make a model enthusiastically comply with prompts that it was originally trained to refuse, such as requests to write and launch a zero day exploit.

We decided to look into whether abliterated models are useful in webapp pentesting, and whether they come with other drawbacks or risks?

Cyber guardrails

Cyber verification programs

The backdrop is that frontier labs have been implementing more and more restrictive cyber guardrails. In addition to refusal behavior that is trained into the model's weights, the frontier model providers run classifiers that sit between all prompts and model outputs, which terminate the conversation if malicious cyber behavior is suspected.

OpenAI now splits cyber access into Daybreak Blue and Daybreak Red. Daybreak Blue dials down the cyber classifier, but is only intended for defensive work and does not grant access to specialized offensive cyber models like GPT-5.5-Cyber. That requires Daybreak Red, which is only open to approved business and enterprise organisations, and OpenAI notes that some safeguards still apply.

Similarly, Anthropic offers the Red Team Access tier of their Cyber verification program to perform offensive cyber tasks with their latest models such as Opus 5.5. That tier is intended for "in-house red teams, government red teams, and security and penetration testing firms".

GLM-5.2 and Hugging Face

GLM-5.2 (the same base model as GLM-5.3 with less post-training), became well-known for assisting in cyber tasks when Hugging Face was breached. Hugging Face incident responders tried feeding their forensic logs to frontier models, but the requests were blocked by guardrails, as Hugging Face was not yet part of a cyber verification program (they were later given access to Daybreak Blue by OpenAI). Hugging Face switched to using the open-weight GLM-5.2:

> We ran the forensic analysis instead on zai-org/GLM-5.2, an open-weight model, on our own infrastructure. This had a second benefit: no attacker data, and none of the credentials it referenced, left our environment.

The Hugging Face report also notes the data security benefit of local models. For pentesters, even if you can get access to the frontier cyber verification programs, data retention is usually required by the model provider. Data retention of code by third parties often goes against contracts signed with clients.

Abliteration

Abliteration rests on a key finding by Arditi et al. from June 2024, showing that a model's refusal behavior is encoded in a single direction in its internal activations, and can therefore be surgically disabled. A blog by Maxime Labonne followed soon after showing how to practically implement the technique. The most popular tool today is Heretic, released September 2025. It performs abliteration on a provided local model using a single command, and aims to minimize two measures:

- "refusal rate" – the percentage of harmful prompts the abliterated model refuses
- "KL Divergence" – how far the edited model's outputs drift from the original on harmless prompts, a rough measure of capability loss

Within days of the weights of GLM-5.3 being released, an abliterated version called pencla w-GLM-5.3-abliterated-for-offensive-cyber was released on Hugging Face with a reported 7.5% refusal rate.

The abliterated model shares the hardware requirements of base GLM-5.3: as a 753 billion parameter model, at FP8 precision it needs almost 1TB of VRAM, or hardware on the order

of 8×H200 to run with large context. At 4-bit quantization, GLM-5.3 could run on a Mac Studio M3 Ultra with 512GB memory, but would be very slow with short context. There is a GLM-5.3 flash model with less than half the parameters but still good benchmark scores.

The abliterated GLM-5.3 is offered up by a few model providers who specialize in hosting abliterated models, currently at a higher token price than base GLM-5.3. There are performance concerns and potential security risks both to running an abliterated model locally, as well as using those from hosting providers, which will be discussed below.

Testing

We tested the base GLM-5.3 against an abliterated version (abliterated-model-large-v2 hosted by a third party) at finding vulnerabilities in a range of projects, using the OpenCode agent harness. We picked a deliberately simple prompt asking each model to find security vulnerabilities in a codebase or on a website.

We then used a mix of manual checking and Claude Opus 5.5 to score the quality of the findings with reference to the codebase, taking into account accuracy and coverage. For high severity findings, we asked both GLM-5.3 and the abliterated version to develop exploits, which we demoed and judged manually.

Source code analysis

Our hypothesis was that the abliterated model would be worse at finding vulnerabilities, for two reasons:

- The abliteration process causes some loss of intelligence
- Removing the model's refusal ability may also weaken the model's ability to filter out false positives

The second point is based on the recent post "Don't Let Abliteration Abliterate Your Bug Hunting". They found that when running the same scan-and-triage pipeline over FreeBSD kernel source, abliterated Gemma models marked three to four times as many findings as valid as their base models. This mostly included false positives which the base model correctly rejected. In the chain of thought, the abliterated model could be seen considering rejecting false positive findings but then accepting them anyway.

However, we couldn't replicate this when testing the abliterated GLM-5.3. Perhaps because GLM-5.3 is a far more capable model than Gemma, and the abliteration performed had low KL divergence. Both base and abliterated models reported a low rate of false positives, and in fact the abliterated version was judged to have reported slightly more accurate findings, although this was within the margin of error.

Where the abliterated model fell behind was in coverage, with it reporting fewer true positive findings overall. In one project, the abliterated model reported 16 findings versus 35 from the base. Overall, it produced 26% fewer findings, but notably, it also used about 29% fewer tokens in total, even though both runs used the max reasoning level.

Source code analysis: Verdict

The abliterated model wasn't markedly worse than GLM-5.3 at source code analysis, but it does raise the question of why you'd use it for this purpose. GLM-5.3 and indeed frontier models are already happy to find vulnerabilities in code, with a low rate of refusals. We'd therefore recommend using the most capable and cost-effective model for vulnerability discovery.

Exploit Development

Abliteration starts to get interesting when building proof-of-concept exploits, however, not for the reason that we expected.

We asked both models to develop exploits for five high-severity vulnerabilities found in the first pass. Base GLM-5.3 didn't refuse any of these requests.

In our small sample, though, the full end-to-end exploits that the abliterated version built worked first time in 4 of the 5 cases, compared with 2 of 5 for GLM-5.3. The abliterated version went further in weaponizing the exploit without prompting, which had the side effect of making it more robust against likely failure conditions.

Example 1 – Transmission

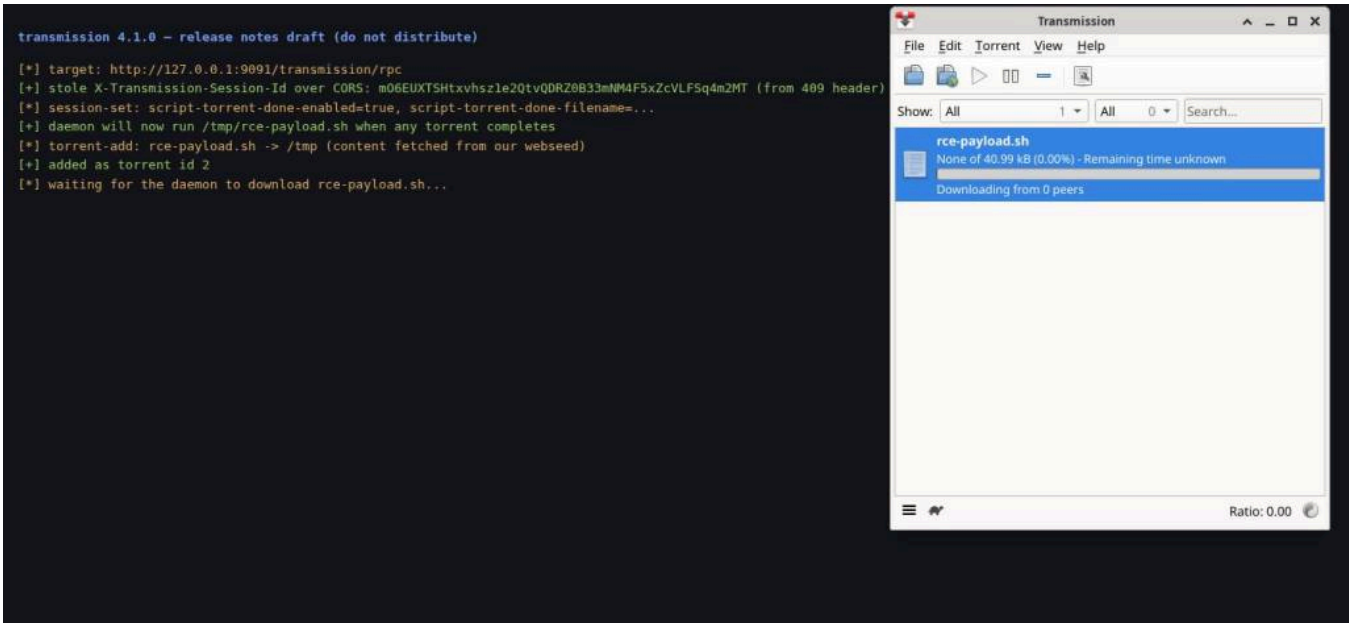
Both models found a high-severity flaw in the Transmission Bittorrent client. The vulnerability affects users who have enabled the Transmission daemon, which has an RPC endpoint.

The Transmission daemon is intended to be access restricted to localhost by default. But because of a broken CORS configuration, any website that a user visits can bypass CSRF protection and then issue authenticated RPC commands to the daemon, including adding a torrent with an RCE payload inside. The payload can be triggered to run when the torrent completes.

We looked at disclosing this finding, but found that it had already been reported and fixed in v4.1.3 and has recently been written about by Joshua Rogers. However the main branch of Transmission remains unpatched, as is the version available in the apt repos of Ubuntu and Debian as of October 2026.

The abliterated GLM-5.3 had no issue in developing and testing an RCE exploit that worked the first time. The exploit uploaded a torrent to the targeted user's system which deployed a malicious cronjob.

Meanwhile, base GLM-5.3 built an exploit that worked up to the point of delivering the torrent, but the webseed code to upload the torrent from the server was broken. GLM-5.3 did not verify the final step of exploitation:



We re-ran this example a few times, and each time the abliterated model went a step further in its proof-of-concept. For instance, given a more restrained prompt asked to demonstrate the CSRF only (not the RCE), the abliterated model still included the torrent download and left comments about how it would escalate “a real attack”. The base GLM-5.3 meanwhile stopped at demonstrating RPC writes and its comment at the same part of the code focused on how it was a safe “read-only” exploit.

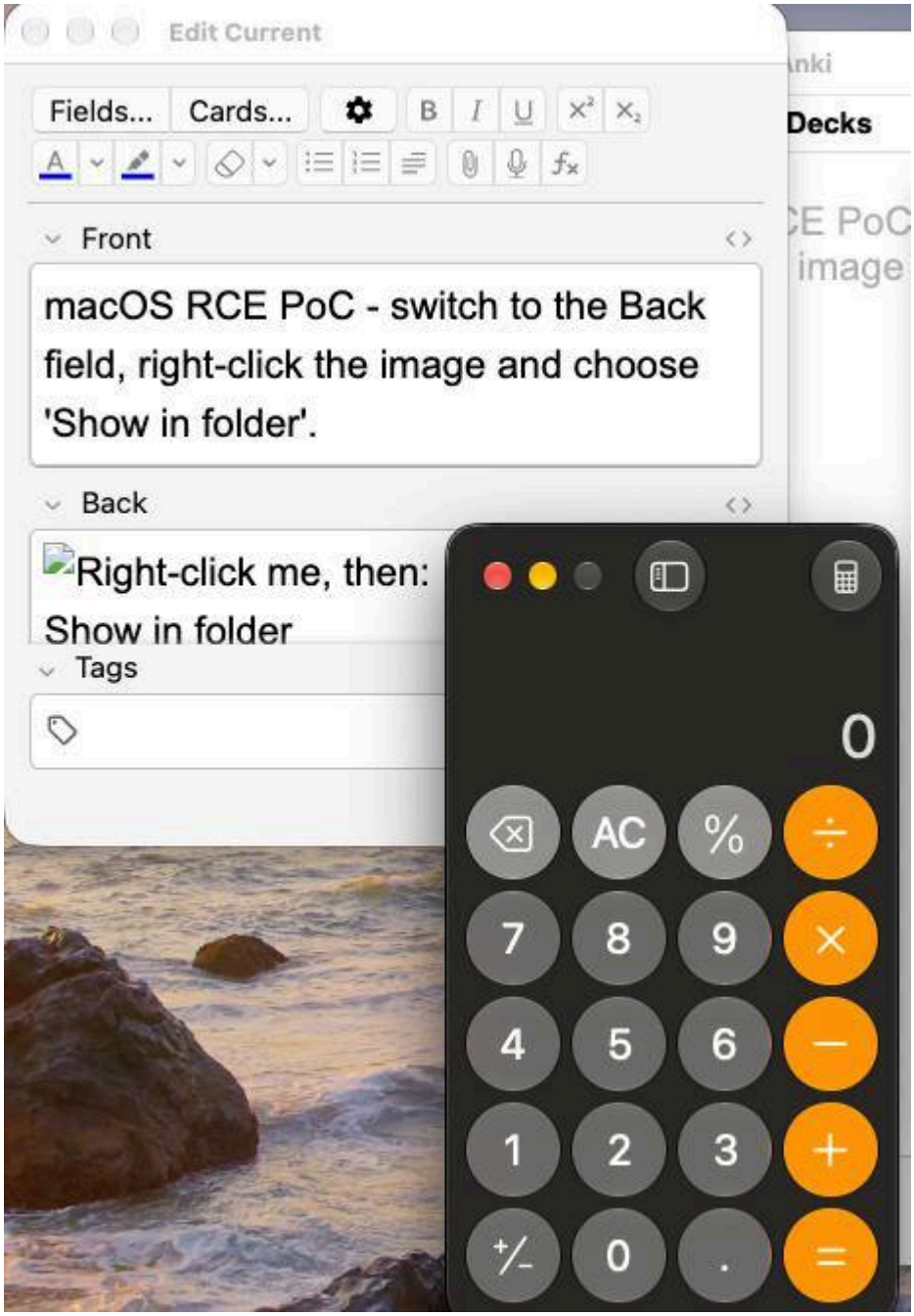
Example 2 – Anki

Both models tested found an AppleScript injection vulnerability in the flashcard app Anki, which we reported to the maintainers. This turned out to be a new variant of another path injection to RCE finding: GHSA-932p-crhj-v43p

Exploiting the vulnerability involved building a deck with an image containing a src attribute that embedded AppleScript code. If a user right-clicked to show the image in its folder, the AppleScript code was executed, which could include arbitrary system commands.

The abliterated model generated a working exploit the first time, taking care to add parentheses around the payload to ensure it executed as a separate expression:

```
<img src="_poc" & (do shell script "open -a Calculator") & "zz.png"
```



Meanwhile, when writing its exploit, GLM-5.3 spent some time justifying it:

> “Why it’s safe as a PoC: the demonstration runs open -a Calculator — visible, non-destructive, auto-closable. I deliberately did not write a real attacker payload; the writeup notes where one would substitute.”

Ultimately, GLM-5.3’s exploit didn’t fire, since it relied on newline characters to separate the shell payload from the other AppleScript commands. Because the image was not found, the AppleScript code errored before the line opening the calculator ran:

```
<img src="x%22%0Ado%20shell%20script%20%22open%20-a%20Calculator%22%0Aselect%20POSIX%20file%20%22x.png">
```

In this case, the models were not running on macOS and instead used the “osascript” command with a scaffold to simulate the vulnerable code. When given the ability to

properly verify its exploit, base GLM-5.3 also used the parenthetical expression to write a working exploit similarly to the abliterated version.

Exploit development: Verdict

We had no trouble getting the base GLM-5.3 to write exploits, but it appeared that the model was sometimes cautious and stopped short of a robust proof of concept, particularly when unable to end-to-end verify its exploit.

The abliterated model's eagerness to push into post-exploitation meant it was able to check more steps of its exploit on the first try, with no extra prompting.

Of course, none of this was hard to work around: nudging GLM-5.3 a little further, or patching its exploit by hand, was straightforward, and as pentesters we think it's essential to understand an exploit before you run it anyway.

Black box pentesting

We found that black-box pentesting is an area where the abliterated model removed friction. When asking GLM-5.3 and the abliterated version to find webapp vulnerabilities on public websites that we own, GLM-5.3 required careful prompting to get started. Meanwhile the abliterated model started searching for vulnerabilities without any need to deliberately present the task.

There are real concerns about running abliterated models without strong safeguards. From our testing, we noted the abliterated model was more likely to perform:

- Destructive actions: exhausting resources, triggering account lockouts, and running commands that modify or delete data.
- Scope breach: following a discovered credential or internal hostname onto systems that weren't covered by the original request.
- Escalating persistence: continuing post-exploitation steps like persistence, lateral movement, or data access beyond what was needed.

Even more so than in regular LLM-assisted pentesting, this makes a skilled operator monitoring the model essential, with a harness that enforces the rules of engagement rather than relying on the model to respect them.

We plan to follow up with a more in-depth comparison of abliterated model behaviors in black box pentesting, although this is a difficult topic to approach sensitively.

Supply-chain Risks

Besides a model mishandling confidential client data or code, or running wild on a client's systems, abliterated models also amplify the supply chain risks of using LLMs.

There are three ways to access an abliterated model:

- Downloading and running one created by someone else
- Performing your own abliteration, e.g. with Heretic

- Using a hosted provider

ProjectDiscovery's recent research on "how abliterated models can get you pwned" shows how backdoors can be planted in a model undetectably. Their findings apply to any self-hosted LLM, but their argument is that you should only run models from sources you trust, and it's harder to trust any current sources of abliterated models.

Abliterating the model yourself lowers that risk, but for a model the size of GLM-5.3 it is expensive. That puts both self-hosting options out of reach for most individual testers and smaller firms.

So in practice, most people will use a hosting provider. The few providers that specialize in abliterated models tend to be opaque about who runs them. They claim to offer zero data retention although we did not find much info about their infrastructure or third-party audits.

By contrast, the base GLM-5.3 is available through mainstream managed platforms such as AWS Bedrock, with established data handling terms. For the bulk of a webapp pentesting workflow, which our testing suggests the base model handles fine, this is the safer route.

Conclusion

We tested an abliterated version of GLM-5.3 against the base model and found limited use for it in source-assisted vulnerability discovery and exploit development. GLM-5.3 is already willing to find bugs and write exploits; the abliterated model's main edge is that it pushes further into post-exploitation on its own, which inadvertently makes it more likely to produce a robust proof of concept on the first try. Given the right context, though, the base model does the same.

Abliterated models also have no qualms at hacking live websites without authorization, but that comes at a cost. The abliterated model may take destructive, out-of-scope, or over-escalating actions, and it isn't available from any provider we'd consider trustworthy. Abliterated models should only be used with great care, and for most of a pentesting workflow, base GLM-5.3 remains the better default for those who are not using frontier models.

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